

Равномерное распределение и тригонометрическая динамика

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Теоретический минимум.

Опр. A sequence $(x_n) \subset [0, 1)$ is *equidistributed* if $\frac{1}{N} \#\{n \leq N : x_n \in [a, b)\} \rightarrow b - a$ for every $0 \leq a < b \leq 1$.

Теорема 1 (критерий Вейля 1916) (x_n) is equidistributed modulo 1 if and only if $\frac{1}{N} \sum_{n \leq N} e^{2\pi i k x_n} \rightarrow 0$ for every nonzero integer k . In particular $(\{n\alpha\})$ is equidistributed for every irrational α .

Теорема 2 (Риман–Лебег и осреднение) If g is periodic with period b and integrable, then $\int_0^b f(x)g(nx) dx \rightarrow \frac{1}{b} \int_0^b f \int_0^b g$ for every continuous f : fast oscillation is replaced by the mean value.

Факт 1 The rotation $x \mapsto x + \alpha$ of the circle with irrational α is uniquely ergodic; the doubling map $\theta \mapsto 2\theta$ is not, and its «тригонометрическая тень» $x \mapsto 2x^2 - 1$ on $[-1, 1]$ preserves the arcsine measure $\frac{dx}{\pi\sqrt{1-x^2}}$.

1. Prove Kronecker's theorem ($\{n\alpha\}$ is dense for irrational α) by the pigeonhole argument, then prove Weyl's criterion and deduce equidistribution.
2. Prove that the sequence $\sin n$ has no limit, that $\sup_n \sin n = 1$, and that the set of its limit points is the whole segment $[-1, 1]$.
3. (IMC) Let $f \in C[0, b]$ and let g be continuous and periodic with period b . Prove Theorem 2, and then compute $\lim_{n \rightarrow \infty} \int_0^\pi \frac{\sin x}{1 + 3 \cos^2(2nx)} dx$.
4. (Мехмар, 2001) Prove that the series $\sum_{n=1}^\infty \sin(2^n x)$ converges only at the points $x = \pi 2^{-k}$, $k \in \mathbb{Z}$.
5. Let $x_1 \in (0, \pi)$ and $x_{n+1} = \sin x_n$. Prove that $x_n \rightarrow 0$ and $x_n \sim \sqrt{3/n}$; find the second term of the asymptotics.